

Leibnitz Theorem

01) y' of $y = \frac{x^2 e^x}{x}$

$$4c_0 u^4 v^0 + 4c_1 u^3 v^1 + 4c_2 u^2 v^2 + 4c_3 u^1 v^3 + 4c_4 u^0 v^4$$

$$u^0 = e^x$$

$$v^0 = x^2$$

$$u^1 = e^x$$

$$v^1 = 2x$$

$$u^2 = e^x$$

$$v^2 = 2$$

$$u^3 = e^x$$

$$v^3 = 0$$

$$u^4 = e^x$$

$$v^4 = 0$$

$$1(e^x)(x^2) + 4(e^x)(2x) + 6(e^x)(2) + 4(e^x)(0) + 0$$

$$= x^2 e^x + 8x e^x + 12 e^x$$

$$= e^x (x^2 + 8x + 12)$$

02) y' of $y = x^2 \ln x$

$$u^0 = \ln x$$

$$v^0 = x^2$$

$$x^{-1}$$

$$u^1 = \frac{1}{x}$$

$$v^1 = 2x$$

$$-(x)^{-2} = -\frac{1}{x^2}$$

$$u^2 = -\frac{1}{x^2}$$

$$v^2 = 2$$

$$+2(x)^{-3}$$

$$u^3 = \frac{2}{x^3}$$

$$v^3 = 0$$

$$-6(x)^{-4}$$

$$u^4 = -\frac{6}{x^4}$$

$$v^4 = 0$$

$$24(x)^{-5}$$

$$u^5 = \frac{24}{x^5}$$

$$v^5 = 0$$

$$= 1\left(-\frac{24}{x^5}\right)(x^2) + 5\left(-\frac{6}{x^4}\right)(2x) + 10\left(\frac{2}{x^3}\right)(2) + 0 + 0 + 0$$

$$= \frac{-24}{x^3} + \frac{-66}{x^3} + \frac{40}{x^3}$$

$$= \frac{4}{x^3}$$

8) y' of $y = x \sin x$

$$u^0 = \sin x \quad v^0 = x$$

$$u^1 = \cos x \quad v^1 = 1$$

$$u^2 = -\sin x \quad v^2 = 0$$

$$u^3 = -\cos x \quad v^3 = 0$$

$$u^4 = \sin x \quad v^4 = 0$$

$$= 1(\sin)(x) + 4(-\cos)(1) + 0 + 0 + 0$$

$$= x \sin x - 4 \cos x$$

Curvature and Radius

Cartesian form $(y=x)$

$$r = \frac{[1 + (y')^2]^{\frac{3}{2}}}{|y''|}$$

Parametric $x(t), y(t)$

$$r = \frac{[(x')^2 + (y')^2]^{\frac{3}{2}}}{|x'y'' - y'x''|}$$

Polar r, θ

$$r = \frac{[r^2 + (r')^2]^{\frac{3}{2}}}{|r^2 + 2(r')^2 - rr''|}$$

1) $x = a(t - \sin t)$
 $y = a(t - \cos t)$

$$x' = a(1 - \cos t)$$

$$y' = a(0 - (-\sin t))$$

$$x'' = a(0 - (-\sin t))$$

$$y' = a \sin t$$

$$x'' = a(\sin t)$$

$$y'' = a \cos t$$

$$r = \frac{[(x')^2 + (y')^2]^{\frac{3}{2}}}{|x'y'' - y'x''|}$$

$$\begin{aligned} r_{\text{num}} &= [(a(1 - \cos t))^2 + (a \sin t)^2]^{\frac{3}{2}} \\ &= [a^2(1 - \cos t)^2 + a^2 \sin^2 t]^{\frac{3}{2}} \\ &= [a^2(1^2 - 2\cos t + \cos^2 t) + a^2 \sin^2 t]^{\frac{3}{2}} \\ &= [a^2 - a^2 2\cos t + a^2 \cos^2 t + a^2 \sin^2 t]^{\frac{3}{2}} \\ &= [a^2(1 - 2\cos t + \sin^2 t + \cos^2 t)]^{\frac{3}{2}} \\ &= [a^2(1 - 2\cos t + 1)]^{\frac{3}{2}} \\ &= [a^2(2 - 2\cos t)]^{\frac{3}{2}} \\ &= a^3(2 - 2\cos t)^{\frac{3}{2}} \\ &= a^3 2^{\frac{3}{2}}(1 - \cos t)^{\frac{3}{2}} \end{aligned}$$

$$\begin{aligned} r_{\text{denom}} &= |(0 - a \cos t)(a \cos t) - (a \sin t)(a \sin t)| \\ &= |a^2 \cos t - a^2 \cos^2 t - a^2 \sin^2 t| \\ &= |a^2(-\cos t + \cos^2 t + \sin^2 t)| \\ &= |-a^2(-\cos t + 1)| \\ &= |a^2 \cos t - a^2| \\ &= |-a^2(-\cos t + 1)| \\ &= a^2(1 - \cos t) \end{aligned}$$

$$= \frac{a \sqrt{2} (1 - \cos \theta)^{3/2}}{\cancel{a^2} (1 - \cos \theta)}$$

$$= 2\sqrt{2} a (1 - \cos \theta)^2$$

8) $x = a (\theta + \sin \theta)$,
 $y = a (1 - \cos \theta)$

$$x' = a (\theta + \sin \theta)$$

$$x' = a (1 + \cos \theta)$$

$$x'' = a (-\sin \theta)$$

$$y' = a (0 - (-\sin \theta))$$

$$y' = a \sin \theta$$

$$y'' = a \cos \theta$$

$$r = \frac{[(x')^2 + (y')^2]^{3/2}}{|x'y'' - y'x''|}$$

$$\frac{[(a + a \cos \theta)^2 + (a \sin \theta)^2]^{3/2}}{|(a + a \cos \theta)(a \cos \theta) - (a \sin \theta)(a - \sin \theta)|}$$

$$= \frac{[a^2 + a^2 \cos^2 \theta + a^2 \sin^2 \theta]^{3/2}}{a^2 \cos \theta + a^2 \cos^3 \theta + a^2 \sin^2 \theta}$$

$$= \frac{[a^2 (1 + 1)]^{3/2}}{a^2 (\cos \theta + 1)}$$

$$= 2\sqrt{2}$$

$$x = a \cos^3 \theta$$

$$y = a \sin^3 \theta$$

$$r = \sqrt{(x')^2 + (y')^2}^{1/2}$$

$$|x''_y - y''_x|$$

$$x' = a \times 3 \cos^2 \theta (-\sin \theta)$$

$$x' = -3a \cos^2 \theta \sin \theta$$

$$x'' = -3a \sin \theta \cos^2 \theta$$

$$x''' = -3a (\cos^2 \theta \cdot \cos \theta + \sin \theta \cdot 2 \cos \theta \sin \theta)$$

$$x'' = -3a (\cos^3 \theta + 2 \sin^2 \theta \cos \theta)$$

$$x'' = -3a \cos^3 \theta - 6a \sin^2 \theta \cos \theta$$

$$y' = 3a \sin^2 \theta \cos \theta$$

$$y'' = 3a (\sin^2 \theta (-\sin \theta) + \cos \theta \cdot 2 \sin \theta \cos \theta)$$

$$y'' = 3a (-\sin^3 \theta + 2 \sin \theta \cos^2 \theta)$$

$$y'' = -3a \sin^3 \theta + 6a \sin \theta \cos^2 \theta$$

$$\text{Num} = [(-3a \sin \theta \cos^2 \theta)^2 + (3a \sin^2 \theta \cos \theta)^2]^{1/2}$$

$$= [9a^2 \sin^2 \theta \cos^4 \theta + 9a^2 \sin^4 \theta \cos^2 \theta]^{1/2}$$

$$= [9a^2 \sin^2 \theta \cos^2 \theta (\cos^2 \theta + \sin^2 \theta)]^{1/2}$$

$$= [9a^2 \sin^2 \theta \cos^2 \theta (1)]^{1/2}$$

$$= 3a \sin \theta \cos \theta$$

$$\text{Denom} = |(-3a \sin \theta \cos^2 \theta)(-3a \sin^2 \theta + 6a \sin \theta \cos^2 \theta) - (3a \sin^2 \theta \cos \theta)(-3a \cos^3 \theta - 6a \sin^2 \theta \cos \theta)|$$

$$= |-9a^2 \sin^4 \theta \cos^2 \theta - 18a^2 \sin^3 \theta \cos^4 \theta + 9a^2 \sin^2 \theta \cos^5 \theta + 18a^2 \sin^4 \theta \cos^3 \theta|$$

$$= |9a^2 \sin^2 \theta \cos^2 \theta (-\sin^2 \theta - 2 \cos^2 \theta + \cos^3 \theta + 2 \sin^2 \theta)|$$

$$= |9a^2 \sin^2 \theta \cos^2 \theta (\sin^2 \theta + \cos^2 \theta)|$$

$$= 9a^2 \sin^2 \theta \cos^2 \theta$$

$$= \frac{3a \sin \theta \cos \theta}{9a^2 \sin^2 \theta \cos^2 \theta}$$

$$= \frac{3a \sin \theta \cos \theta}{9a^2 \sin^2 \theta \cos^2 \theta}$$

$$\theta) \quad r(1 + \cos \theta) = a$$

$$r = \frac{a}{1 + \cos \theta}$$

$$= \left[(r)^2 + (r')^2 \right]^{3/2}$$

$$1 (r)^3 + 2(r')^2 - r' r''$$

$$r = a (1 + \cos \theta)^{-1}$$

$$r = \frac{a}{1 + \cos \theta}$$

$$r' = a (-1) (1 + \cos \theta)^{-2} (0 - \sin \theta)$$

$$r' = a \sin \theta (1 + \cos \theta)^{-2}$$

$$r' = \frac{a \sin \theta}{(1 + \cos \theta)^2}$$

$$r'' = a \sin \theta (-2) (1 + \cos \theta)^{-3} (0 - \sin \theta)$$

$$r'' = \frac{+2a \sin^2 \theta}{(1 + \cos \theta)^3}$$

$$\text{Num} = \left[\left(\frac{a}{1 + \cos \theta} \right)^2 + \left(\frac{a \sin \theta}{(1 + \cos \theta)^2} \right)^2 \right]^{3/2}$$

$$= \left[\frac{a^2}{(1 + \cos \theta)^2} + \frac{a^2 \sin^2 \theta}{(1 + \cos \theta)^4} \right]^{3/2}$$

$$= \left[\frac{a^2 (1 + \cos \theta)^2 + a^2 \sin^2 \theta}{(1 + \cos \theta)^4} \right]^{3/2}$$

$$= \left[\frac{a^2 (1^2 + 2 \cos \theta + \cos^2 \theta) + a^2 \sin^2 \theta}{(1 + \cos \theta)^4} \right]^{3/2}$$

$$= \left[\frac{a^2 + a^2 2 \cos \theta + a^2 \cos^2 \theta + a^2 \sin^2 \theta}{(1 + \cos \theta)^4} \right]^{3/2}$$

$$= \left[\frac{a^2 (1 + 2 \cos \theta + \cos^2 \theta + \sin^2 \theta)}{(1 + \cos \theta)^4} \right]^{3/2}$$

$$= \left[\frac{a^2 (2 + 2 \cos \theta)}{(1 + \cos \theta)^4} \right]^{3/2}$$

$$= \left[\frac{2a^2 (1 + \cos \theta)}{(1 + \cos \theta)^4} \right]^{3/2}$$

$$= \left[\frac{2a^2}{(1 + \cos \theta)^3} \right]^{3/2}$$

$$= \frac{2\sqrt{2} a^3}{(1 + \cos \theta)^{9/2}}$$

$$\text{Denom} = \left| \left(\frac{a}{1 + \cos \theta} \right)^3 - 2 \left(\frac{a \sin \theta}{(1 + \cos \theta)^2} \right)^2 - \left(\frac{a \sin \theta}{(1 + \cos \theta)^2} \right) \left(\frac{2a \sin^2 \theta}{(1 + \cos \theta)^3} \right) \right|$$

$$= \frac{a^3}{(1 + \cos \theta)^3} - \frac{2a^2 \sin^2 \theta}{(1 + \cos \theta)^4} - \frac{2a \sin^3 \theta}{(1 + \cos \theta)^5}$$

$$= a^3 (1 + \cos \theta)^3$$

Partial Differentiation

$$1) \quad f(x, y) = x^4 + 6\sqrt{y} - 10$$

$$= 4x^3 + 0 + 0$$

$$= 4x^3$$

$$= 0 + 6\left(\frac{1}{2}\right)y^{-\frac{1}{2}} - 0$$

$$= 1 + \frac{3}{\sqrt{y}}$$

$$2) \quad w = x^2y - 10y^2z^3 + 43z - 7\sin(4y)$$

$$\frac{\partial w}{\partial x} = 2xy - 0 + 43 - 0$$

$$= 2xy + 43$$

$$\frac{\partial w}{\partial z} = 0 - 10y^2z^2 + 43 - 0$$

$$= -10y^2z^2 + 43$$

$$\frac{\partial w}{\partial y} = x^2 - 20yz^3 - 7\sec^2(4y)(4)$$

$$= x^2 - 20yz^3 - 28\sec^2(4y)$$

$$3) \quad h = t^7 \ln(s^2) + \frac{9}{t^3} - 7\sqrt{s^4}$$

$$h = t^7 \ln(s^2) + 9t^{-3} - 7s^2$$

$$\frac{\partial h}{\partial t} = 7t^6 \ln(s^2) - 9(3)t^{-4} - 0$$

$$= 7t^6 \ln(s^2) - \frac{27}{t^4}$$

$$\frac{\partial h}{\partial s} = t^7 \frac{1}{s^2} \cdot 2s + 0 - 7(2)s$$

$$= \frac{2t^7}{s} - 14s$$

$$\frac{\partial^2 h}{\partial t \partial s} = 7t^6 \frac{1}{s^2} \cdot 2s - 0$$

$$= \frac{14t^6}{s}$$

$$\frac{\partial^2 h}{\partial s \partial t} = \frac{2(7)t^6}{s} - 0$$

$$= \frac{14t^6}{s}$$